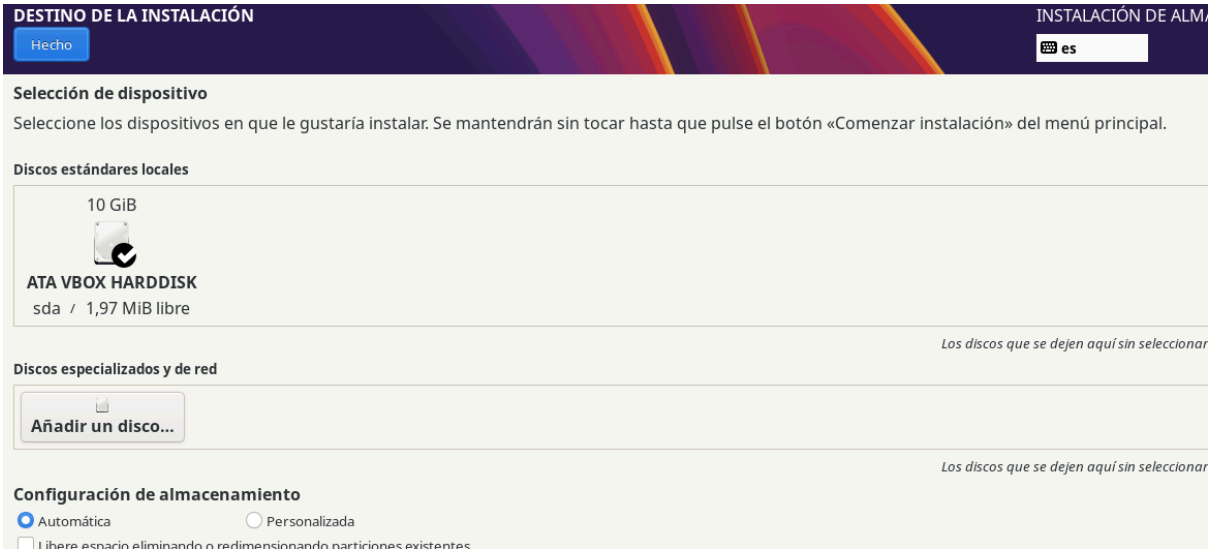
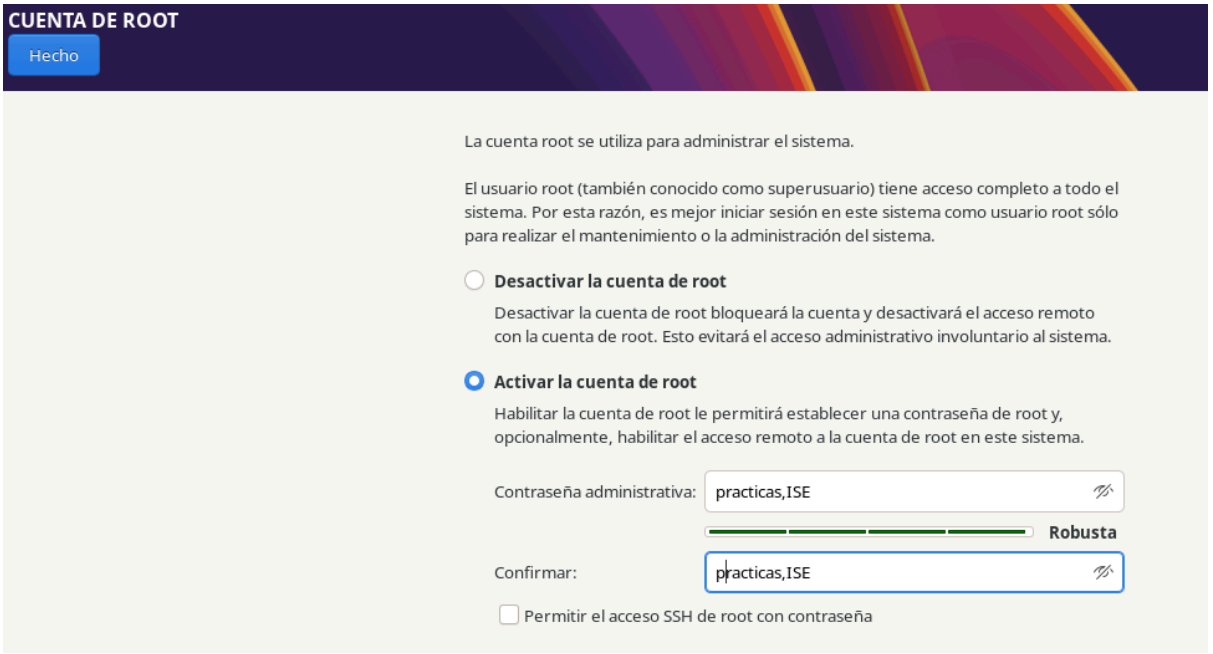
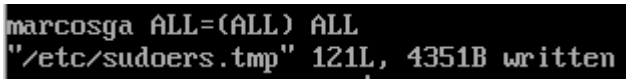
P1_L2

Indicamos el destino de la instalación y configuramos la cuenta de root.





Modifico /etc/sudoers con visudo para otorgar todos los permisos



Comprobamos la configuración por defecto con lsblk

```
AlmaLinux 10.0 (Purple Lion)
Kernel 6.12.0-55.9.1.el10_0.x86_64 on an x86_64

localhost login: marcosga
Password:
[marcosga@localhost ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   10G  0 disk
├─sda1                8:1    0    1M  0 part
├─sda2                8:2    0    1G  0 part /boot
└─sda3                8:3    0    9G  0 part
  ├─almalinux-root 253:0    0    8G  0 lvm  /
  └─almalinux-swap 253:1    0    1G  0 lvm  [SWAP]
sr0                  11:0    1 1024M  0 rom
[marcosga@localhost ~]$
```

Añadimos nuestro sdb y comprobamos que se haya añadido correctamente

```
[marcosga@localhost ~]$ lsblk | grep disk
sda                   8:0    0   10G  0 disk
sdb                   8:16   0    5G  0 disk
```

Utilizamos el siguiente comando para realizar una partición en sdb

```
[marcosga@localhost ~]$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x70d5bc64.

Command (m for help):
```

Creamos la partición con n, p indica que es de tipo primaria, y lo comprobamos viendo la tabla de particiones

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-10485759, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485759, default 10485759):

Created a new partition 1 of type 'Linux' and of size 5 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x70d5bc64

Device     Boot Start      End  Sectors Size Id Type
/dev/sdb1        2048 10485759 10483712   5G 83 Linux
```

Creamos el volumen físico con pvcreate y lo comprobamos con pvs

```
[marcosga@localhost ~]$ sudo pvcreate /dev/sdb1
[sudo] password for marcosga:
  Physical volume "/dev/sdb1" successfully created.
[marcosga@localhost ~]$ pvdisplay
  WARNING: Running as a non-root user. Functionality may be unavailable.
  /run/lock/lvm/P_global:aux: open failed: Permission denied
[marcosga@localhost ~]$ sudo pvs
  PV         VG        Fmt  Attr PSize  PFree
  /dev/sda3  almalinux lvm2 a--  <9.00g     0
  /dev/sdb1            lvm2 ---  <5.00g <5.00g
```

Creamos otro volumen físico y lo comprobamos

```
[marcosga@localhost ~]$ sudo vgs
[sudo] password for marcosga:
  VG        #PV #LV #SN Attr   VSize  VFree
  almalinux   1   2   0 wz--n- <9.00g    0
[marcosga@localhost ~]$ sudo vgextend almalinux /dev/sdb1
  Volume group "almalinux" successfully extended
[marcosga@localhost ~]$ sudo vgs
  VG        #PV #LV #SN Attr   VSize  VFree
  almalinux   2   2   0 wz--n- 13.99g <5.00g
```

Comprobamos los volúmenes lógicos, añadimos el nuevo volumen lógico llamado new_var con el comando lvcreate con 3 GB en nuestro volumen físico llamado almalinux

```
[marcosga@localhost ~]$ sudo lvs
  LV   VG        Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root almalinux -wi-ao---- <8.00g
  swap almalinux -wi-ao----  1.00g
[marcosga@localhost ~]$ sudo lvcreate -n new_var -L 3G almalinux
  Logical volume "new_var" created.
[marcosga@localhost ~]$ sudo lvs
  LV     VG        Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  new_var almalinux -wi-a-----  3.00g
  root    almalinux -wi-ao---- <8.00g
  swap    almalinux -wi-ao----  1.00g
```

Creamos un filesystem para el volumen lógico

```
[marcosga@localhost ~]$ sudo mkdir /new_var
[marcosga@localhost ~]$ sudo mkfs -t ext4 /dev/almalinux/new_var
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: 184db279-930c-418a-943a-1714b7e7c785
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Accedemos al volumen lógico usando el comando mount

```
[marcosga@localhost ~]$ sudo mount /dev/almalinux/new_var /new_var
[marcosga@localhost ~]$ mount
/dev/mapper/almalinux-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4096k,nr_inodes=214051,mode=
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,pt
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsd
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=349300k,nr_inodes=819200,mod
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=35,pgrp=1,timeout=
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabe
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,seclabel,pagesi
tmpfs on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexe
oswap)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
/dev/sda2 on /boot type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=3
tmpfs on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,rela
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,seclabel,size=174648k,n
/dev/mapper/almalinux-new_var on /new_var type ext4 (rw,relatime,seclabel)
```

Entramos en modo root y vemos con el comando systemctl status que estamos en modo mantenimiento

```
[root@localhost ~]# systemctl status
● localhost.localdomain
    State: maintenance
    Units: 338 loaded (incl. loaded aliases)
     Jobs: 0 queued
   Failed: 0 units
    Since: Sat 2025-09-27 12:55:04 CEST; 49min ago
  systemd: 257-9.el10_0.1.alma.1-g1905557
   CGroup: /
           ├─init.scope
           │ └─1 /usr/lib/systemd/systemd --switched-root --system --deserialize=47
           ├─system.slice
           │ ├─crond.service
           │ │ └─1405 /usr/sbin/anacron -s
           │ ├─rescue.service
           │ │ ├─1601 /usr/lib/systemd/systemd-sulogin-shell rescue
           │ │ ├─1602 /usr/sbin/sulogin
           │ │ ├─1603 bash
           │ │ ├─1630 systemctl status
           │ │ └─1631 less
           │ ├─systemd-journald.service
           │ │ └─595 /usr/lib/systemd/systemd-journald
           │ └─systemd-udevd.service
           │   └─udev
           │     └─639 /usr/lib/systemd/systemd-udevd
           └─user.slice
             └─user-1000.slice
               └─user@1000.service
                 └─init.scope
                   ├─1307 /usr/lib/systemd/systemd --user
                   └─1309 "(sd-pam)"
```

Copiamos la información desde /var actual hasta el volumen lógico y lo comprobamos con el comando ls -laZ

```
[root@localhost ~]# cp -a /var/. /new_var
[root@localhost ~]# ls -laZ /var
total 16
drwxr-xr-x. 19 root root system_u:object_r:var_t:s0           4096 Sep 27 12:51 .
dr-xr-xr-x. 19 root root system_u:object_r:root_t:s0           250 Sep 27 13:29 ..
-rw-r--r--.  1 root root system_u:object_r:etc_runtime_t:s0     208 Sep 27 12:48 .updated
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 adm
drwxr-xr-x.  8 root root system_u:object_r:var_t:s0             88 Sep 27 12:48 cache
drwxr-xr-x.  2 root root system_u:object_r:kdump_crash_t:s0      6 Mar  6  2025 crash
drwxr-xr-x.  3 root root system_u:object_r:system_db_t:s0       18 Sep 27 12:48 db
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 empty
drwxr-xr-x.  2 root root system_u:object_r:public_content_t:s0   6 Oct 29  2024 ftp
drwxr-xr-x.  2 root root system_u:object_r:games_data_t:s0       6 Oct 29  2024 games
drwxr-xr-x.  3 root root system_u:object_r:var_t:s0             18 Sep 27 12:48 kerberos
drwxr-xr-x. 20 root root system_u:object_r:var_lib_t:s0       4096 Sep 27 12:51 lib
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 local
lrwxrwxrwx.  1 root root system_u:object_r:var_lock_t:s0        11 Sep 27 12:48 lock -> ../run/lock
drwxr-xr-x.  7 root root system_u:object_r:var_log_t:s0       4096 Sep 27 12:51 log
lrwxrwxrwx.  1 root root system_u:object_r:mail_spool_t:s0      10 Oct 29  2024 mail -> spool/mail
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 nis
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 opt
drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 preserve
lrwxrwxrwx.  1 root root system_u:object_r:var_run_t:s0          6 Sep 27 12:48 run -> ../run
drwxr-xr-x.  6 root root system_u:object_r:var_spool_t:s0       56 Sep 27 12:48 spool
drwxrwxrwt.  2 root root system_u:object_r:tmp_t:s0              6 Sep 27 13:39 tmp
drwxr-xr-x.  2 root root system_u:object_r:var_yp_t:s0           6 Oct 29  2024 yp
[root@localhost ~]#
```

Entramos en vi y añadimos esta línea de código y guardamos, luego hacemos mount -a y comprobamos con lsblk para ver que se ha añadido correctamente

```
/dev/mapper/almalinux-new_var /var ext4 defaults 0 0
```

```
[root@localhost ~]# lsblk
NAME                   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                      8:0    0   10G  0 disk
├─sda1                   8:1    0    1M  0 part
├─sda2                   8:2    0    1G  0 part /boot
└─sda3                   8:3    0    9G  0 part
  ├─almalinux-root     253:0    0    8G  0 lvm  /
  └─almalinux-swap     253:1    0    1G  0 lvm  [SWAP]
sdb                      8:16   0    5G  0 disk
└─sdb1                   8:17   0    5G  0 part
  └─almalinux-new_var  253:2    0    3G  0 lvm  /var
```

Con ls / vemos que estan creadas /var y /var_old

```
[marcosga@localhost ~]$ ls /
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  new_var  opt  proc  root  run  sbin  srv  sys  tmp  usr  var  var_old
```

Liberamos el espacio del antiguo var con el comando umount -a y mv /var /var_old y lo vemos con ls -laZ

```
[marcosga@localhost ~]$ ls -laZls -laZ /var_old/
total 16
4 drwxr-xr-x. 19 root root system_u:object_r:var_t:s0            4096 Sep 27 12:51 .
0 dr-xr-xr-x. 20 root root system_u:object_r:root_t:s0            265 Oct  6 08:45 ..
4 -rw-r--r--.  1 root root system_u:object_r:etc_runtime_t:s0     208 Sep 27 12:48 .updated
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 adm
0 drwxr-xr-x.  8 root root system_u:object_r:var_t:s0             88 Sep 27 12:48 cache
0 drwxr-xr-x.  2 root root system_u:object_r:kdump_crash_t:s0      6 Mar  6  2025 crash
0 drwxr-xr-x.  3 root root system_u:object_r:system_db_t:s0       18 Sep 27 12:48 db
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 empty
0 drwxr-xr-x.  2 root root system_u:object_r:public_content_t:s0   6 Oct 29  2024 ftp
0 drwxr-xr-x.  2 root root system_u:object_r:games_data_t:s0       6 Oct 29  2024 games
0 drwxr-xr-x.  3 root root system_u:object_r:var_t:s0             18 Sep 27 12:48 kerberos
4 drwxr-xr-x. 20 root root system_u:object_r:var_lib_t:s0       4096 Sep 27 12:51 lib
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 local
0 lrwxrwxrwx.  1 root root system_u:object_r:var_lock_t:s0        11 Sep 27 12:48 lock -> ../run/lock
4 drwxr-xr-x.  7 root root system_u:object_r:var_log_t:s0       4096 Sep 27 12:51 log
0 lrwxrwxrwx.  1 root root system_u:object_r:mail_spool_t:s0      10 Oct 29  2024 mail -> spool/mail
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 nis
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 opt
0 drwxr-xr-x.  2 root root system_u:object_r:var_t:s0              6 Oct 29  2024 preserve
0 lrwxrwxrwx.  1 root root system_u:object_r:var_run_t:s0          6 Sep 27 12:48 run -> ../run
0 drwxr-xr-x.  6 root root system_u:object_r:var_spool_t:s0       56 Sep 27 12:48 spool
0 drwxrwxrwt.  2 root root system_u:object_r:tmp_t:s0              6 Sep 27 13:39 tmp
0 drwxr-xr-x.  2 root root system_u:object_r:var_yp_t:s0           6 Oct 29  2024 yp
[marcosga@localhost ~]$
```